

# A machine-checked proof that the maximum number of stable matchings of order five is sixteen

Jiarui Xu\*

August 31, 2026

## Abstract

Let  $f(n)$  be the maximum number of stable matchings of a stable-marriage instance with  $n$  men and  $n$  women; determining  $f$  is a problem of Knuth (1976) and Open Problem #1 of Gusfield and Irving (1989). The value  $f(5) = 16$  was announced by Eilers (2022, OEIS A357269) on the strength of an unpublished constraint-programming search that produced no checkable artifact. We give the first machine-checkable proof. A Lean 4 development defines the problem in  $\sim 40$  lines, proves—with no `sorry` and only the three standard Mathlib axioms—the theorem

$$\text{f5\_eq\_16\_of\_unsat} : \left( \forall \rho \in P_{120}, \neg \text{Sat}(\Phi_\rho) \right) \longrightarrow \left( \forall I, \text{WF}(I) \rightarrow \text{sc}(I) \leq 16 \right) \wedge \left( \exists I, \text{WF}(I) \wedge \text{sc}(I) = 16 \right),$$

where the 120 propositional formulas  $\Phi_\rho$  are themselves defined in Lean and printed verbatim to DIMACS by a 30-line exporter. Each  $\Phi_\rho$  is refuted by `kissat`, each proof is validated by `drat-trim`, and each resulting LRAT certificate is checked end-to-end by the formally verified checker `cake_lpr`: 120/120 verified, about two hours on a laptop. The architecture follows the empty-hexagon development of Subercaseaux et al. (ITP 2024). As by-products we independently confirm Eilers’ companion counts (OEIS A344669): there are exactly 507,254,400 maximal profiles of order 5 (176,130 up to symmetry), obtained by enumerating all 4,227,120 canonical solutions.

## 1 Introduction

A stable-marriage instance of order  $n$  equips  $n$  men and  $n$  women each with a strict preference order over the opposite side; a perfect matching is *stable* if no man and woman prefer each other to their assigned partners. Gale and Shapley proved stable matchings always exist. Let  $f(n)$  denote the maximum number of stable matchings over instances of order  $n$ . Knuth posed the behaviour of  $f$  as a research problem in 1976 [1]; Gusfield and Irving made it Open Problem #1 of their monograph [2]. Asymptotically  $2.28^n \leq f(n) \leq 3.55^n$  [3, 4, 5], and exactly (OEIS A357269 [6]):

$$f(1), \dots, f(5) = 1, 2, 3, 10, 16.$$

The last and hardest value,  $f(5) = 16$ , was announced by Dan Eilers in September 2022, computed with a MiniZinc constraint-programming model. The result is trustworthy—we confirm

---

\*`jxucoder@gmail.com`. The formal development, computations and drafting were carried out with the assistance of Claude (Anthropic); every numerical claim comes from executed, logged runs in the accompanying repository, and every formal claim is checked by the Lean kernel.

it here by independent means—but it had, until now, no proof object: constraint solvers do not emit certificates, the search is not reproducible from any published artifact, and the paper announced alongside the OEIS entry has not appeared. For contrast, the best human-provable upper bound at  $n = 5$  is  $3.55^5 \approx 555$ .

## Contributions.

1. **A machine-checked proof of  $f(5) = 16$**  in the architecture of [7]: a Lean 4 development (zero `sorry`s; axioms `propext`, `Classical.choice`, `Quot.sound`) reduces the theorem to the unsatisfiability of 120 concrete CNF formulas that are *defined inside Lean* and printed verbatim for the solvers; the 120 refutations are checked by the formally verified proof checker `cake_lpr` [9]. The kernel-checked part comprises a symmetry reduction, a case-covering lemma, and the encoding-faithfulness theorem; the lower bound is a kernel-computed evaluation of an explicit witness.
2. **Independent confirmations** of every unreplicated number in this corner of the OEIS:  $f(5) = 16$  itself, and the maximal-profile counts  $A_{344669}(3, 4, 5) = 1092, 144, 507, 254, 400$  together with Eilers’ reduced count 176,130, via exhaustive enumeration with a different toolchain.
3. **A complete verification artifact**: a repository with a step-by-step third-party verification guide (`VERIFYING.md`), pinned toolchain versions and hashes, a differential-testing harness binding the formal definitions to an independently validated reference implementation, and continuous integration that rebuilds the development and checks the axiom base on every commit.

## 2 The Lean development

**Definitions.** An instance is a pair of  $5 \times 5$  rank tables (`mrank` $[m][w]$  = position of woman  $w$  in man  $m$ ’s order, 0 best; `wrank` symmetric); well-formedness `WF` requires both tables to have five rows, each a permutation of  $\{0, \dots, 4\}$ . A matching is one of the 120 permutations  $\mu$ ; it is stable iff no pair  $(m, w)$  with  $w \neq \mu(m)$  has both `mrank` $[m][w] < \text{mrank}[m][\mu(m)]$  and `wrank` $[w][m] < \text{wrank}[w][\mu^{-1}(w)]$ . The count `sc`( $I$ ) filters Mathlib’s `List.permutations`. These definitions total about forty lines and are the entire human-review surface of the formal claim.

**Upper bound.** The kernel-checked chain has three stages. *Symmetry* (`reduce_man0`): relabelling women by man 0’s rank row fixes man 0’s row to the identity, preserves well-formedness, and preserves `sc`; the proof constructs the relabelling bijection on matchings and carries stability across it pointwise. *Covering* (`cube_covering`): for well-formed  $I$ , man 1’s rank row lies in the 120-element list  $P_{120} = \text{permutations}[0, \dots, 4]$ . *Faithfulness* (`cube_faithful`): for each  $\rho \in P_{120}$ , any well-formed  $I$  with man 0 identity, man 1 row  $\rho$ , and `sc`( $I$ )  $\geq 17$  yields an assignment satisfying the CNF  $\Phi_\rho$  of Section 3 — the assignment reads the preference variables off  $I$  and points seventeen selector slots at the first seventeen stable matchings of  $I$  in enumeration order; each clause family is then discharged. Chaining the three: if every  $\Phi_\rho$  is unsatisfiable then `sc`( $I$ )  $\leq 16$  for all well-formed  $I$ .

**Lower bound.** An explicit witness instance (found by SAT in 0.1s) is embedded, and `sc(witnessI) = 16` is proved by kernel computation. Mathlib’s `permutations` uses well-founded recursion, which the kernel cannot evaluate, so the count is transported across `List.permutations.perm_permutations'` to the structurally recursive `permutations'` and evaluated there by `decide`. A standalone variant (`Witness.lean`) proves the same count from first principles using only `propext`.

**Main theorem.** `f5.eq_16_of_unsat` (statement in the abstract) combines both directions; `#print axioms` reports exactly `[propext, Classical.choice, Quot.sound]`, re-checked by CI on every commit.

### 3 The certificates

**Encoding.**  $\Phi_\rho$  uses pairwise preference variables with transitivity (forbidding both 3-cycles on every triple), unit clauses pinning man 0 to the identity and man 1 to  $\rho$ , and a *selector* block: variables  $y_{t,\mu}$  ( $t < 17$ ,  $\mu$  indexing  $P_{120}$ ) with (i) each slot nonempty, (ii) consecutive slots strictly increasing in  $\mu$ -index (killing slot symmetry and forcing seventeen distinct selections), and (iii) selected matchings admitting no blocking pair. Sizes: 2,140 variables, 157,197 clauses. The encoding was chosen to make faithfulness formalizable by hand; no cardinality-network semantics enter the proof. As a positive control, the 16-slot variant of the cube containing the witness is satisfiable, and the decoded model recounts to a valid 16-matching instance.

**Pipeline.** The formulas are printed from the Lean definitions by a 30-line exporter. Each cube is refuted by `kissat 4.0.4` (hardest cube 115s — the one containing the extremal instances), each DRAT proof is validated and elaborated to LRAT by `drat-trim`, and each LRAT is checked by `cake_lpr`, whose soundness is itself machine-checked down to machine code [9]. Result: **120/120 s VERIFIED UNSAT**, about two hours end-to-end on an Apple M4 Max (8-way parallel). Outside the Lean kernel the trust base is `cake_lpr` and the 30-line printer, and nothing else — the solvers are checked, not trusted.

### 4 Cross-validation

Four independent anchors tie the formal definitions to reality: (1) a reference counter validated against the extremal Latin instances of orders 3, 4, 5, 6 recorded in OEIS A351413 (3/10/9/48, all reproduced); (2) a differential test: on 300 random instances the Lean definitions and the reference counter agree exactly; (3) the witness chain: SAT model  $\rightarrow$  reference recount  $\rightarrow$  Lean kernel count, all 16; (4) enumeration: blocking-clause enumeration over all 120 cubes yields exactly 4,227,120 canonical (man-0 = id) profiles with 16 stable matchings; multiplying by the 5! woman-relabellings gives 507,254,400 = A344669(5), and dividing by the 4! man-relabellings gives 176,130, Eilers’ reduced count — both previously unreplicated. The same runs confirm A344669(3) = 1092 (full brute force over all 46,656 profiles) and A344669(4) = 144 (six canonical solutions).

Before adopting the present architecture we also refuted “ $\geq 17$ ” monolithically three times (different encodings and solver configurations, including a pure-RUP `kissat -plain` proof of 2.1GB verified by `drat-trim`, and a Mathlib `lrat_proof` kernel import of the analogous  $n=4$  certificate, which reproves  $f(4) = 10$  inside Lean in ten seconds).

## 5 Related work

Our architecture — verified encoding in Lean, certificates checked by `cake_lpr`, unsatisfiability entering the main theorem as an explicit hypothesis — is that of the empty hexagon number verification [7]. The end-to-end Keller-conjecture verification [8] has since internalized even the checker into Lean by reflection; migrating our certificate step onto such a checker (or LRAT-Catcher [12]) is mechanical future work. Other certified computational values include  $R(4, 5) = 25$  [10], Schur number five, the packing chromatic number of the square grid, and  $K_8(4, 2) = 23$  [11]. On the stable-matching side the computational lineage is Eilers’ ( $f(4)$ ,  $f(5)$ , the profile counts) building on Thurber [3] and ultimately Knuth [1]; the asymptotic bounds are [4, 5].

## 6 Conclusion

$f(5) = 16$  now has a proof whose every step is checked either by the Lean kernel or by a formally verified certificate checker, with a forty-line human-review surface. The first open case is  $f(6)$ : the best lower bound is 48 (a dihedral Latin instance), conjectured exact (OEIS A357271). The search space grows by eleven orders of magnitude, so a direct extension will require either much stronger symmetry breaking inside the certificates or a structural route through rotation posets — for which the 176,130 extremal instances enumerated here are the natural microscope.

**Artifact.** Repository with code, proofs, logs, verification guide (`VERIFYING.md`) and CI; certificate archive with DOI to accompany the final version. Reproduction cost:  $\sim 20$  minutes for the Lean development,  $\sim 2\text{--}3$  hours for the certificates.

**Acknowledgments.** The priority for  $f(5) = 16$  and for the A344669 counts belongs to Dan Eilers; this work confirms his announcements and supplies the proof objects. We thank the OEIS, and the authors of `kissat`, `drat-trim` and `cake_lpr`.

## References

- [1] D. E. Knuth, *Mariages stables*, Presses de l’Université de Montréal, 1976.
- [2] D. Gusfield and R. W. Irving, *The Stable Marriage Problem: Structure and Algorithms*, MIT Press, 1989.
- [3] E. G. Thurber, Concerning the maximum number of stable matchings in the stable marriage problem, *Discrete Math.* 248 (2002), 195–219.
- [4] A. R. Karlin, S. Oveis Gharan and R. Weber, A simply exponential upper bound on the maximum number of stable matchings, *STOC 2018*.
- [5] C. Palmer and D. Pálvölgyi, At most  $3.55^n$  stable matchings, *FOCS 2021*.
- [6] OEIS, sequences A357269, A357271, A344669, A351413, <https://oeis.org>.
- [7] B. Subercaseaux, W. Nawrocki, J. Gallicchio, C. Codel, M. Carneiro and M. J. H. Heule, Formal verification of the empty hexagon number, *ITP 2024*; arXiv:2403.17370.

- [8] J. Gallicchio, C. Codel, J. Avigad and M. J. H. Heule, An end-to-end verification of Keller’s conjecture, *ITP 2026*.
- [9] Y. K. Tan, M. J. H. Heule and M. O. Myreen, cake\_lpr: verified propagation redundancy checking in CakeML, *TACAS 2021*.
- [10] T. Gauthier and C. E. Brown, A formal proof of  $R(4, 5) = 25$ , arXiv:2404.01761, 2024.
- [11] A Lean-certified proof of  $K_8(4, 2) = 23$ , arXiv:2606.16688, 2026.
- [12] LRAT-Catcher: importing SAT solver certificates into Lean 4 by reflection, arXiv:2607.00815, 2026.